

ANIMAL BIRTHDAY PREDICTOR (ABP)

Final Implementation & AI Agent Development Documentation

Purpose: Complete the existing Flutter + Supabase ABP application without unnecessarily restructuring the existing database or breaking existing data.

IMPORTANT DATABASE STATUS: The two previously supplied database SQL files have already been run successfully in the Supabase SQL Editor. The AI agent must treat those files as already executed and must NOT ask the user to rerun them.

1. Mandatory Database Change Policy

From this point onward, every NEW database change must be delivered as a separate incremental SQL file. The user should be able to open that file and run it directly in the Supabase SQL Editor.

- First inspect the existing schema, tables, columns, constraints, foreign keys, indexes and RLS policies.
- Reuse the existing schema whenever possible.
- Do not recreate or rerun the two original SQL files.
- Do not unnecessarily rename, drop, recreate or restructure existing tables or columns.
- If a database change is genuinely required, create a separate SQL migration file for that change.
- The migration must be independently runnable from the Supabase SQL Editor.
- Prefer additive, non-destructive changes that preserve all existing data.
- Clearly explain why the migration is needed and what it changes.
- After the user runs the migration, verify that the application code matches the updated schema.

Required SQL file naming:

```
abp_incremental_migration_001.sql
```

Future migrations should use 002, 003, etc. Do not mix new SQL changes into unrelated application files and do not ask the user to manually reconstruct SQL commands.

If no schema change is needed: explicitly state that the existing database schema is sufficient and do not create an unnecessary migration.

2. Universal Animal Architecture

Keep the universal animal registry architecture. One real animal should have one animal record, referenced by ID from breeding, pregnancy, foal and other records. Horse and Dog are required now, while the architecture should remain extensible for future Cat/Other modules.

3. Existing Animal vs Add New

Wherever an animal reference is needed, provide both options where appropriate:

```
SELECT EXISTING ANIMAL OR ADD NEW ANIMAL
```

Select Existing opens the user's saved animals, imports already-known basic fields and returns the existing animal ID. Add New opens a blank animal form, creates the new animal and returns its ID to the current process.

Critical: tapping a saved animal must open that animal's details/edit page. It must never open a blank Add Animal form. Only an explicit Add action opens a new blank form.

4. Horse Module

Complete and preserve the approved Horse workflow: Animal Details, Mare, Breeding, Recipient, Pregnancy, Scans, Preventative Care, Foal, Markings, Contacts, CRUD and Certificate.

Breeding Style must be a single-select/radio field and must remain linked wherever the same breeding-method field is shown.

Stallion remains free text, but an optional existing-animal selection must also be available. Sire and Dam must support existing/new selection where an animal reference is required.

5. Pregnancy Date Engine

Method	Base	Scan 1	Scan 2	Scan 3	Due
Natural	Mare cover	+14d	+30d	+45d	+341d
Chilled	Mare cover	+14d	+30d	+45d	+341d
Frozen	Mare cover	+14d	+30d	+45d	+340d
Embryo Transfer / ICSI	Recipient transfer	+7d	+30d	+45d	+332d

Calculate dates once when the breeding/recipient record is saved and persist them. Never silently recalculate on read, because a vet-adjusted date must not be overwritten. Manual editing of calculated dates must persist.

6. Contacts Directory

Provide a proper directory for Vet, Farrier and Dentist. From a record, the user can either import an existing directory contact or add a new contact directly. Any contact added directly from a record must also be automatically saved to the main Contacts Directory for future reuse. The directory should also allow contacts to be added manually.

7. Markings

Markings are part of the relevant Mare/Recipient/Foal/Animal record. Provide three image slots: Left Side, Right Side and Head View. Every image field supports Camera and Gallery. Markings should open Camera first, with Gallery as a secondary option.

8. Foal Module

Implement Foal Details with photo, name, date of birth, linked Mare/Dam, Stallion, optional Recipient, breed, sex, IGG, notes, status, microchip, DNA, gelded status/date and Stud Book/Breeding Association. Markings and Preventative Care must be accessible from the Foal record.

Foal status: Sold, Keep, Transferred. Sold/Transferred must prompt for new owner information such as name, phone and address. Keep requires no additional ownership information.

9. Dog / Puppy Module

Implement the Dog/Puppy module according to the supplied Puppy reference images. Do not simply rename the Horse UI.

- Puppy Identification: Puppy Name/ID, Sex, Colour, Birth Order, Microchip Number, Collar/Tag Colour.
- Puppy Photo: Take Photo and Choose from Gallery.
- Weights: Birth Weight and ongoing weights with date, age in days, weight and notes.
- Health & Treatments: Worming, Vaccinations, Vet/Health Checks and Microchip.
- Next Due Reminder for worming, vaccination and vet check.
- Going Home Information: Date Going Home, New Owner/Home, Contact Number and Notes.
- General Notes, Timeline, Dogs Home, Next Puppy and Preventative Care navigation.
- Dog/Puppy data must persist to Supabase and support Edit/Delete.

Dog preventative care must not be forced into the Horse preventative-care structure. Dog uses Date Given + Date Due pairs per item, while the approved Horse layout uses a single date plus done state.

10. Certificates — Horse and Dog

Use one reusable certificate architecture with species-specific terminology. Certificates are auto-populated from existing records and are read-only. The user should see a clear certificate preview and be able to export it as PDF.

- Horse/Foal Certificate.

- Dog/Puppy Certificate.
- Auto-populated from saved records.
- Read-only certificate view.
- Clear preview page.
- PDF export.
- Do not make the user re-enter information that already exists in ABP.
- Certificate should be a polished take-home summary, not a complete database dump.

11. CRUD — Edit and Delete

Every applicable user-created record must have working Edit and Delete functionality. This includes animals, breeding records, pregnancy records/scans, preventative care, markings, foals, puppies, puppy weights/health records, going-home information and contacts.

Edit must update the real Supabase record. Delete must ask for confirmation, delete the correct record, refresh the UI, handle relationships safely and never affect another user's data.

12. User Data Isolation

Only the authenticated user's data may be displayed or modified. All database operations must respect Supabase RLS and ownership. Do not rely only on client-side filtering for security.

13. Responsive UI

The entire application must be responsive. Test approximately 360dp, 390–430dp and 768dp+ widths in portrait and landscape. Avoid hardcoded screen dimensions. Long names, notes and contact details must wrap or truncate safely. No text clipping, overlapping, broken layouts or unintended horizontal overflow. Content-heavy screens must scroll correctly. Keyboard opening must not break layouts.

14. Dashboard & Navigation

Audit the current Dashboard and navigation. Existing animal cards/list items must open existing details. Explicit Add actions must open creation forms. Fix any current Dashboard flow that incorrectly opens new-animal forms or otherwise loses the existing record context.

15. Required AI Agent Workflow

- Audit the existing Flutter project before changing code.
- Audit navigation, models, repositories, providers and Supabase integration.
- Inspect the current database schema and RLS before proposing SQL changes.
- Treat the two original SQL files as already executed.
- Do not ask the user to rerun the original SQL files.
- Compare the current implementation against this documentation.
- Reuse working code and components wherever possible.
- Implement/fix Horse and Dog/Puppy workflows.
- Implement reusable Select Existing / Add New animal flow.
- Implement Edit/Delete and user-specific data isolation.
- Implement auto-populated read-only certificates and PDF export.
- Fix Dashboard and existing-animal navigation.
- Test persistence, images, reminders and responsive UI.
- If a schema change is required, create a separate SQL migration file.

- Tell the user exactly what the new SQL migration changes and that it must be run in Supabase SQL Editor.
- Verify the app against the migrated schema after the user executes it.

16. Mandatory SQL Migration Deliverable

Whenever new database changes are genuinely required, the agent **MUST** create a separate file such as:

```
abp_incremental_migration_001.sql
```

That file must contain the complete SQL needed for the new change and must be directly runnable in Supabase SQL Editor. Do not make the user manually copy SQL statements from the chat.

The agent must explicitly report: (1) why the change is needed, (2) which tables/columns/policies are affected, (3) that the two original SQL files have already been executed and must not be rerun, and (4) the exact new SQL file that should be run.

17. Final Acceptance Checklist

- Saved animals open their existing details/edit page.
- Add Animal opens a blank form only through an explicit Add action.
- Select Existing and Add New work wherever applicable.
- Mare, Recipient, Sire and Dam flows work.
- Stallion supports free text plus optional existing-animal selection.
- Breeding Style is single-select.
- Pregnancy calculations are correct and editable.
- Calculated dates persist and are not silently overwritten.
- Reminders work.
- Contacts Directory works and direct-created contacts are saved for reuse.
- Markings support Left, Right and Head images with Camera/Gallery.
- Foal Sold/Keep/Transferred workflow works.
- Dog/Puppy module follows the supplied Puppy flow.
- Dog/Puppy data persists correctly.
- Horse and Dog/Puppy certificates are auto-populated and read-only.
- Certificate preview and PDF export work.
- Edit and Delete update the real database.
- User data is isolated correctly.
- Dashboard navigation is correct.
- No unnecessary schema restructuring occurs.
- Any genuinely required schema change is delivered as a separate SQL migration file.
- Responsive UI works without overflow, clipping or broken text.

END OF ABP IMPLEMENTATION DOCUMENTATION